Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation

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***Abstract*—Large language models let non-player characters hold conversations, but they forget early facts, propose changes that break game rules, and cost more as context grows. We present OBSIDIAN FLASK, a dialogue engine that separates generation from state control: a dual-tier memory whose retrieval is scoped to the current location and character, and an event-sourced world state in which every proposed change must pass a symbolic validation barrier before commit. We compare it with six memory baselines at an equal budget, including the retrieval rule of the base paper, Generative Agents, in a memory-by-control factorial with seven ablations, on seven worlds and four models, and audit every committed change with an invariant oracle that shares no code with the validator. With a 7-billion-parameter backbone, it recalled 83.0% of facts 20 or more turns old on held-out worlds, against 0.0% for a rolling context and 63.0% for the base paper's rule, but not more than flat vector memory (84.3%) or its earlier version (88.6%). Scoping raised mean reciprocal rank from 0.742 to 0.910. No output committed a state violation through the barrier, which rejected 0.7% of lawful live requests, and recovery was exact after all 300 simulated crashes. These results are preliminary.**

***Index Terms*—Large language models, non-player characters, long-term memory, event sourcing, runtime verification.**

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Source code, benchmark and run outputs: https://github.com/parthmital/RAG-Driven-NPC-Narrative-Engine.